

Team Name: Net-runners

Project Title: Graph Neural Networks (GNNs) - Predicting Molecular Energy Gaps

Project Summary: Scientists currently use a method called Density Functional Theory (DFT) to calculate the HOMO-LUMO gap of molecules, which is a key property for understanding chemical reactivity. While DFT is very accurate, it is computationally expensive and takes hours to simulate just a single molecule. We want to train a Graph Neural Network (GNN) to predict this gap in significant lesser time. This is interesting because accelerating these calculations can help researchers discover new materials for solar panels, batteries, and medicine much faster than before. Our project uses the PCQM4Mv2 dataset, which contains over 4 million molecules, providing a significant challenge for scaling deep learning models on non-Euclidean graph data.

Implementation Approach: We will implement a Deep Neural Network using the PyTorch Geometric library. Our main objective is to build and train a GNN that can predict the energy gap with a Mean Absolute Error (MAE) of less than 0.12 eV. We will specifically implement and analyze a Virtual Node in a global node that connects to every atom in a molecule to see if it helps information travel better across large molecular graphs compared to a standard GNN. For our implementation, we will write our own custom model layers, training loops, and evaluation scripts rather than just running existing OGB example code. We plan to run experiments comparing a baseline Graph Convolutional Network (GCN) against our improved version that includes Skip Connections to handle vanishing gradients and Batch Normalization to stabilize the training of 3.7 million samples.

Resources / Related Work & Papers: We are basing our work on the GIN (Graph Isomorphism Network) paper, which explains the best way for atoms to share information with their neighbors. We are also following the OGB-LSC paper, which sets the original rules and benchmarks for this dataset. While top models like Graphormer use complex Transformers, we chose the Virtual Node approach because it is a faster way to help the model see the whole molecule at once. We will use PyTorch Geometric for the graph math and RDKit to turn molecules into data. Our research will also look at over-smoothing, a common problem where atoms start to look identical if the model has too many layers.

Datasets: We are using the **PCQM4Mv2** dataset from the Open Graph Benchmark.

- **Link:** <https://ogb.stanford.edu/docs/lsc/pcqm4mv2/>

- **Description:** This dataset contains 4,379,310 molecules. Each is represented as a graph where atoms are nodes with 9 features and bonds are edges with 3 features.

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